

Machine, Data and Learning

Assignment 1

13 January 2026

Instructions

- The deadline for the assignment is **January 20, 2026**.
 - Submissions made after the deadline will be penalized by **20%** of the total marks. Late submissions will only be accepted up to **January 23, 2026**.
 - You are expected to submit a **handwritten solution** to Moodle.
 - The submission should be named as: `<RollNo> Assignment1.pdf`
 - **Total Marks: 20**
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1. Use the truth tables method to determine whether the following expression is a tautology:
(2M)

$$(p \rightarrow \neg q) \vee (\neg p \rightarrow q)$$

2. Let $\psi : (q \vee \neg p) \rightarrow (q \vee p)$ and $\phi = q$.

a. Show by truth table method that $\phi \Rightarrow \psi$. (2M)

b. Show $\phi \Rightarrow \psi$ using a set of valid arguments (axioms/rules of inference). (2M)

3. We want to check if a given assignment of integers to a 9×9 grid is a valid Sudoku puzzle.

Let x_{ijk} denote a proposition. It is true if the cell in row i , column j contains integer k , and is false otherwise.

Write down the set of axioms that x_{ijk} should satisfy to be a valid Sudoku solution. (6M)

4. Convert the following formula into Conjunctive Normal Form (CNF) and Disjunctive Normal Form (DNF): (4M)

$$(\neg p \rightarrow q) \rightarrow r$$

5. Analyze the following inductive argument regarding the color of horses: (4M)

Premise 1: $n = 1$: Within any group of 1 horses, all horses are of the same color.

Premise 2: $n = k$: Within any group of any k horses, all horses are of the same color. Therefore, horses 1 to k are of the same color. Similarly, horses 2 to $k + 1$ are of the same

color. Note that horses 2 to k belong to both the groups. Therefore, horses of both group are of the same color.

Conclusion : $n = k + 1$: Within any group of $k + 1$ horses, all horses are of the same color.

Is **Conclusion** a valid tautology or a fallacy? Describe the reasoning behind your answer.